TEST3-QUESTION BANK (COMPUTER SCIENCE) CLASS: XII

1. Consider the following table EMP containing information about an employee: -

EMPNO	ENAME	DEPT	SALARY
1	ALEX	MUSIC	60000
2	PETER	ART	67000
3	JHONY	WE	55000
4	RAMBO	P&HE	48000

Following python code is written to access the records of the table EMP. What will be the output of the following code:

```
query = “Select * from emp”
mycursor.execute(query)
result1= mycursor.fetchone()
result1= mycursor.fetchone()
result1= mycursor.fetchone()
d= int(result1[3])
print(d*3)
```

2. Consider the following Python code written to access the record of CODE passes to function. Complete the missing statements:

```
def Search(eno):
    query=”select * from emp where emp where empno = _____”.format(eno)
    mycursor .execute(query)
    result = mycursor. _____
    print(results)
```

3. Consider the following python code to display all records from table: EMP

```
def ShowAll():
    query=”select * from emp”
    mycursor .execute(query)
    result = mycursor. fetchall()
    for results in row:
        print(results)
```

But query is giving error, what could be the possible reason?

4. Explain the purpose of the following lines in the given code: -

```
import mysql.connection as con
mycon = con.connect(host=”localhost”, user=”root”,passwd= “admin”,
database=”employee”)
mycursor=mycon.cursor()                    ---------- 1
mycursor.execute(“select * from emp”)          ---------- 2
mydata = mycursor.fetchall()               ---------- 3
nrec = mycursor.rowcount()                 ---------- 4
print(“Total records fetched till now are:”,nrec)
```

5.Differentiate between fetchone() and fetchmany() ?

6. What is the role of a cursor? Write syntax to create a cursor?

7.

PNo	NameAge		Department	DateofAdm	Charges	Sex
1	Mayank	65	Surgery	23/02/2018	600	M
2	Babita	24	ENT	01/01/2019	400	F
3	Kashish	45	Orthopaedic	19/12/2018	400	M
4	Tarun	12	Surgery	01/10/2018	600	M
5	Manisha	36	ENT	12/01/2018	400	F
6	Imran	16	ENT	24/02/2018	400	M
7	Ankita	NULL	Cardiology	20/08/2018	800	F
8	Zoya	45	Gynecology	22/02/2018	500	F
9	Kush	19	Cardiology	13/01/2019	800	M
10	Shalini	31	Medicine	19/02/2018	300	F

Note: PNo is the primary key in the above table.

Write SQL commands for the statements (a) to (g) on the table HOSPITAL.

(a) To show all the information about the patients of the cardiology department.

(b) To list the names of female patients who are either in the orthopaedic or surgery department.

(c) To list the name of all the patients with their date of admission in ascending order.

(d) To display the patient's name, charges, the age for female patients only.

(e) To count the number of patients with age > 30.

(f) To display various departments.

(g) To display the number of patients in each department.

(h) To display the details of all the patients whose name starts with the alphabet 'Z'.

(i) To change the age of the patient Kush to 20.

(j) To increase the charges of all the patients by 5%.

(k) To remove the record of the patient whose Name is Tarun.

(l) To add another column DocName(Doctor Name) of the type varchar in the above table.

(m) To display patient detail whose age is missing(null).

(n) To decrease the charges by 5% of all the patients admitted to the ENT department.

(o) To insert a new row in the HOSPITAL table with the following data: 11,' Kasif', 37,'ENT','2018-02-25', 300, 'M'.

(p) To set charges to NULL for all the patients in the Surgery department.

(q) To display patient details who are giving charges in the range 300 and 400 (both inclusive).

(r) To display the details of that patient whose name second character contains 'a'.

(s) To display total charges of ENT Department.

(t) To display details of the patients who admitted in the year 2019.

(u) To display the structure of the table hospital.

(v) Write the command to create the above table

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8.

Write SQL commands for the following: Table: TEACHER

TID	NAME	AGE	DEPT	DATEOFJOIN	SAL	SEX
T118	Navin	40	Computer	2010-01-10	12000	M
T107	Chetna	37	History	2008-03-24	20000	F
T105	Sandeep	46	Maths	2006-12-12	30000	M
T110	Sangeeta	35	History	2010-07-01	25000	F
T101	Rudransh	42	Maths	2004-09-05	40000	M
T121	Neeraj	38	Physics	2011-04-01	28000	M

(a) To show information about the teachers of the history department.
(b) To list the names of teachers earning a salary between 20000 and 30000.
(c) To count the number of male teachers.
(d) Display gender wise total number of teachers.
(e) To list the name and age of teachers of female teachers in descending order of date of join.
(f) Increase the salary by 10% for Maths departments.
(g) To delete the record of teacher Neeraj.

9.

Observe the following STUDENTS and EVENTS tables carefully and write the name of the RDBMS operation which will be used to produce the output as shown in LIST ? Also, find the Degree and Cardinality of the LIST.

STUDENTS

NO	NAME
1	Tara Mani
2	Jaya Sarkar
3	Tarini Trikha

EVENTS

EVENTSCODE	EVENTNAME
1001	Programming
1002	IT Quiz

Output :

LIST

NO	NAME	EVENTCODE	EVENTNAME
1	Tara Mani	1001	Programming
1	Tara Mani	1002	IT Quiz
2	Jaya Sarkar	1001	Programming
2	jaya Sarkar	1002	ITQ Quiz
3	Tarini Trikha	1001	Programming
3	Tarini Trikha	1002	IT Quiz

10.

Write SOL queries for (i) to (iv) and find outputs for SQL queries (v) to (viii) which are based on the tables.

TABLE : CUSTOMERS

NO	CNAME	ADDRESS
101	Richa jain	Delhi
101	Surbhi Sinha	Chennai
103	Lisa Thomas	Bangalore
104	Imran Ali	Delhi
105	Roshan Singh	Chennai

TABLE : TRANSACTION

TRNO	CNO	AMOUNT	TYPE	DOT
T001	101	1500	Credit	2017-11-23
T002	103	2000	Debit	2017-05-12
T003	102	3000	Credit	2017-06-10
T004	103	12000	Credit	2017-09-12
T004	101	1000	Debit	2017-09-05

(a) To display details of all transactions of TYPE Credit from Table TRANSACTION.
(b) To display the CNO and AMOUNT of all transactions done in the month September 2017 from the table TRANSACTION.
(c) To display the last date of transaction (DOT) from the table TRANSACTION for the customer having CNO as 103.
(d) To display all CNO,CNAME and DOT (date of transaction) of those CUSTOMERS from tables CUSTOMERS and TRANSACTION who have done transactions more than or equal to 2000.
(e) Select count(*), avg(amount) from transaction where dot>= '2017-06-01';
(f) Select cno, count(*), max (amount) from transaction group by cno having count(*)> 1;
(g) Select cno, cname from customer where address not in ('Delhi, 'Bangalore');
(h) Select distinct cno from transaction;

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Given the following relation: STUDENT

Table: STUDENT

No.	Name	Age	Department	Dateofadm	Fee	Sex
1	Pankaj	24	Computer	10/01/97	120	M
2	Shalini	21	History	24/03/98	200	F
3	Sanjay	22	Hindi	12/12/96	300	M
4	Sudha	25	History	01/07/99	400	F
5	Rakesh	22	Hindi	05/09/97	250	M
6	Shakeel	30	History	27/06/98	300	M
7	Surya	34	Computer	25/02/97	210	M
8	Shikha	23	Hindi	31/07/97	200	F

Write SQL commands for the following queries

(a) To show all information about the students of History department.
(b) To list the names of female students who are in Hindi department.
(c) To list the names of all students with their date of admission in ascending order.
(d) To display student's name, fee, age for male students only.
(e) To count the number of students with Age>23.

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12.

Write SQL queries for (a) to (d) and find outputs for SQL queries (e) to (i), which are based on the tables.

TABLE: WATCHES

Watchid	Watch_name	Price	Type	Qty_store
W001	High time	10000	Unisex	100
W002	Life time	15000	Ladies	150
W003	Wave	20000	Gents	200
W004	High fashion	7000	Unisex	250
W005	Golden time	25000	Gents	100

TABLE: SALE

Watchid	Qty_Sold	Quarter
W001	10	1
W003	5	1
W002	20	2
W003	10	2
W001	15	3
W002	20	3
W005	10	3
W003	15	4

(a) To display all the details of those watches whose name ends with 'time'
(b) To display watch's name andprice of those watches which have price range between 5000-15000
(c) To display total quantity in store of unisex type watches
(d) To display watch name and their quantity sold in first quarter
(e) Select max(price),min(qty_store) from watches;
(f) Select quarter, sum(qty_sold) from sale group by quarter;
(g) Select watch_name,price,type from watches w, sale s where w.watchid=s.watchid;
(h) Select watch_name, qty_store, sum(qty_sold),qty_store;

13.

Consider the SchoolBus table given below:

Table: School Bus

Rtno	Area_covered	Capacity	Nostuds	Dist	Trans	Charges
1	Vasantkunj	100	120	10	Shivamtravels	100000
2	HauzKhas	80	80	10	Anand travels	85000
3	Pitampura	60	55	30	Anand travels	60000
4	Rohini	100	90	35	Anand travels	100000
5	Yamuna Vihar	50	60	20	Bhalla Co.	58000
6	Krishna Nagar	70	80	30	Yadav Co.	80000
7	Vasundhara	100	110	20	Yadav Co.	100000
8	Paschim Vihar	40	40	20	Speed travels	55000

(a) State the command that will give the output as:

Area_covered
Yamuna Vihar
Krishna Nagar
Vasundhara

(b) What will be the output of the following command?

Select * from schoolbus where distance=20 order by charges;

(b) Ravi has given the following command to obtain the Longest distance

Select max(distance) from schoolbus where group by transporter;

But he is not getting the desired result. Help him by writing the correct code.

(d) Help Saumya to write the command to display the name of the transporter who is having lowest capacity in his schoolbus?

14.In the table “Student”, Priya wanted to increase the Marks (Column Name:Marks) of those students by 5 who have got Marks below 33.
She has entered the following statement:

SELECT Marks+5 FROM Student WHERE Marks<33;

Identify errors (if any) in the above statement. Rewrite the correct SQL statement.

15. Differentiate between cardinality and degree of a table with the help of an example.

16. What is the importance of a Primary Key in a table? Explain with a suitable example.

17. Write SQL statement to add a column “COUNTRY” with data type and size as VARCHAR (70) to the existing table named “PLAYER”. Is it a DDL or DML or TCL command?

18. Write SQL query to display details of all rows except those rows that have CITY as ‘DELHI’ or ‘MUMBAI’ or ‘CHANDIGARH’.

19. How is a database related to a table?

20.How is HAVING clause similar to WHERE clause? How is HAVING clause different from WHERE clause? Explain with the help of examples of each.

21.1 Difference between Drop and Delete Command
21.2 Difference between truncate and Delete Command
21.3 Difference between Order By and Group By
21.4 Difference between DDL command and DML command.
21.5 Difference between CHAR and VARCHAR
21.6 Difference between ALTER and UPDATE command.
21.7 Difference between Primary key and Foreign key
21.8 Difference between Candidate key and Alternate key
21.9 Difference between fetchone() and fetchmany(n)
21.10 What is cursor object?
21.11 What is result set?
21.12 What is close()?
21.13 What is commit()?
21.14 What is rollback()?
21.15 What is connect()?
21.16 Define the following terms:

- Database
- Instance
- Schema
- Metadata
- Tuples
- Attributes